

FORM 2 - COMPLETE SPECIFICATION (ILLUSTRATIVE EXAMPLE)

THE PATENTS ACT, 1970 & THE PATENTS RULES, 2003

A STANDARDIZED BIOACTIVE AYURVEDIC EXTRACT OF WITHANIA SOMNIFERA ENRICHED IN WITHANOLIDES FOR STRESS AND FATIGUE MANAGEMENT

APPLICANT NAME: [Applicant / Organization Name]

INVENTOR: [Inventor Name]

REGULATORY PATHWAY: To be determined based on intended product classification and applicable Indian regulatory requirements

IPC / CPC CLASSIFICATION: Illustrative classification only; final classification requires professional patent review

1. FIELD OF THE INVENTION

The present invention relates to the technical domain of standardized Ayurvedic botanical preparations. Specifically, it describes an illustrative process for obtaining a reproducible *Withania somnifera* (Ashwagandha) root extract with controlled withanolide content and converting the extract into a standardized oral dosage form intended for investigation in stress and fatigue management.

2. PATENTABILITY ANALYSIS: TRADITIONAL KNOWLEDGE & NOVEL PROCESS

Traditional use of Ashwagandha and conventional Ayurvedic preparations are not, by themselves, treated here as the inventive contribution. The proposed inventive concept is instead directed to a reproducible extraction, fractionation, standardization and quality-control process that produces a defined chemical profile. Any actual patentability assessment must be performed against prior art, traditional knowledge databases, and the applicable provisions of Indian patent law.

3. STANDARDIZED EXTRACT PROFILE & VALIDATION

The illustrative standardized fraction is characterized using chromatographic fingerprinting and may be specified by a validated assay for total withanolides. Example quality parameters include: **(i)** defined withanolide range; **(ii)** moisture and ash limits; **(iii)** microbial limits; **(iv)** absence or controlled levels of specified contaminants; and **(v)** batch-to-batch chromatographic similarity.

The numerical limits below are examples for demonstrating document structure and are not clinical or regulatory claims. Any therapeutic efficacy or safety statement must be supported by appropriate preclinical and human studies.

4. PATENT CLAIMS (ILLUSTRATIVE)

1. A standardized botanical extract obtained from *Withania somnifera* root, characterized by a controlled withanolide profile and reproducible chromatographic fingerprint within predefined quality specifications.
2. The standardized extract as claimed in claim 1, formulated as an oral tablet, capsule, granule or liquid dosage form containing a defined quantity of the standardized extract.
3. A process for producing the standardized extract of claim 1, comprising: (a) authentication and cleaning of *Withania somnifera* roots; (b) controlled drying and size reduction; (c) extraction using a selected hydroalcoholic solvent system under controlled temperature and time; (d) filtration and concentration; (e) selective fractionation to reduce unwanted non-target constituents; (f) drying of the enriched fraction; and (g) chromatographic standardization and batch-release testing.
4. The standardized extract as claimed in claim 1, wherein the final product is subjected to identity, assay, microbial, heavy-metal, pesticide-residue and stability testing according to the applicable product standards.
5. The process as claimed in claim 3, wherein raw material traceability is maintained from authenticated botanical source through extraction, fractionation, packaging and final batch release.

5. PROCESS FLOW

- 1 Raw material sourcing & botanical authentication

- 2 Cleaning → drying → controlled milling
- 3 Hydroalcoholic extraction under controlled conditions
- 4 Filtration and concentration
- 5 Selective fractionation / purification
- 6 Drying and conversion to standardized extract
- 7 HPLC/UPLC fingerprint and marker-compound assay
- 8 Safety and quality testing
- 9 Formulation into final Ayurvedic/botanical dosage form
- 10 Stability, packaging, traceability and batch release

6. REGULATORY & STATUTORY CONSIDERATIONS

The regulatory route depends on whether the final product is presented as an Ayurvedic medicine, a phytopharmaceutical, a food/supplement, or another regulated category. Before commercialization, the applicant should determine the applicable licensing authority, clinical-evidence requirements, manufacturing standards, labeling rules, biodiversity-access obligations, and any traditional-knowledge disclosure requirements.

7. ADVANTAGES OF THE PROPOSED PROCESS

- Improves batch-to-batch consistency through chemical standardization.
- Creates a traceable link between authenticated raw material and finished extract.
- Provides measurable quality-control parameters rather than relying only on crude plant identity.
- Can support development of a reproducible oral dosage form after appropriate validation.

IMPORTANT NOTE

This PDF is an illustrative patent-style technical example created in the same general format as the uploaded *Boswellia* phytopharmaceutical document. It is not a substitute for a patent attorney, regulatory consultant, clinical evidence, or an official Indian Patent Office filing. The process parameters, claims, therapeutic indications, and regulatory statements must be independently validated before real-world use.